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MedGraph: Maintaining Patient State with Graph Memory for Long-Term Medical Dialogue

Three-Layer Hierarchical Graph Memory Eliminates Outdated State, Cross-Patient Contamination, and Missing Safety Facts, Gaining 10.5 Points on MedMemoryBench

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Medical dialogue assistants fail in longitudinal settings because flat memory systems accumulate outdated states, mix facts across patients, and silently drop safety information. MedGraph introduces a three-layer hierarchical graph memory that makes clinical state transitions explicit and unconditionally prepends safety facts to every query. On MedMemoryBench it exceeds the strongest of five baselines by 10.4–10.5 percentage points with both GPT and Qwen backbones.

AT A GLANCE

Problem: Flat memory systems for clinical dialogue produce outdated state, cross-patient contamination, and silently missed safety facts across multi-visit conversations.

Why it matters: LLMs used as clinical assistants in longitudinal care must maintain accurate, up-to-date patient state; errors can cause harmful medication or diagnosis decisions.

Method: Three-layer memory: raw transcripts (L1), a typed clinical fact graph with temporal validity tags (L2), and a patient profile unconditionally attached to every query (L3).

Result: 10.4–10.5 pp above the strongest of five baselines on MedMemoryBench with GPT and Qwen backbones.

Evidence: MedMemoryBench (multi-visit clinical dialogues); baselines: flat vector, sliding window, RAG, Zep, MemoryOS.

The Big Picture

LLMs deployed as clinical dialogue assistants need to maintain accurate knowledge of a patient's medical state across many visits over weeks or months. Through error analysis of five baseline systems on longitudinal medical dialogues, the authors identify three recurring failure modes:

1. **Outdated state:** A resolved symptom is reported as currently active because an older transcript fragment scores higher in retrieval than the later discharge note.
2. **Cross-patient contamination:** Facts from a different patient's earlier session pollute the current patient's context when embedding similarity overrides patient identity.
3. **Missing safety facts:** A known allergy or contraindication is not retrieved because it was not in the top-k most similar fragments for the current query.

Why Existing Approaches Fall Short

Flat vector memory embeds all clinical utterances and retrieves top-k by similarity. It does not distinguish whether a fact is active or resolved, and similarity can produce facts from other patients.

Sliding-window context loses long-term facts (a years-old allergy, a chronic condition diagnosed at first visit) when they fall out of the window.

Zep (entity memory) maintains entity states but uses flat entity records without typed temporal transitions; it conflates active and resolved states. **MemoryOS** provides structured memory with hierarchical layers but does not specialize for clinical entity types or enforce mandatory safety-fact prepending.

Core Mechanism: Three-Layer Memory

Layer 1 — Raw dialogues. Full session transcripts stored verbatim, searched by BM25 retrieval for specific historical passages.

Layer 2 — Clinical fact graph. A typed knowledge graph over entities (symptoms, diagnoses, medications, lab results, procedures) with typed relations and temporal validity tags: *active*, *resolved*, or *historical*. Type-specific update rules: if an entity is new, add as active; if matched with a resolution signal, change tag to resolved; if repeated observation with no state change, update timestamp only.

Layer 3 — Patient profile (safety block). Per-patient compact record of allergies, contraindications, chronic conditions, emergency contacts. *Unconditionally prepended* to every query context, bypassing retrieval. Each patient has a fully separate graph instance, making cross-patient contamination architecturally impossible.

Technical Formulation

At query time the context assembled for the LLM backbone is:

```
C = Profile_p ⊕ Graph_p^{active} ⊕ RAG(D, q)
```

where Profile_p is the L3 safety block (mandatory), $\text{Graph_p}^{\text{active}}$ is the subgraph of currently active entities serialized as structured text, and RAG(D, q) is BM25-retrieved transcript fragments. The "resolved" signal is detected by a small classifier trained on clinical resolution phrases.

Learning or Inference Procedure

MedGraph uses no end-to-end learning. Components: a clinical NER/RE pipeline pre-trained on MIMIC-IV notes (entity and relation extraction); a resolution-signal classifier (BERT-based, fine-tuned on clinical resolution annotation); and an arbitrary LLM backbone (tested with GPT-4o-mini and Qwen2.5-72B). All components are frozen at inference time.

What the Guarantee Says

No formal guarantee is claimed. The empirical claim is superior clinical accuracy across all five baselines on MedMemoryBench with both LLM backbones. The error analysis identifies three failure modes and verifies per-category that MedGraph's design eliminates each by construction.

Experimental Findings

- **Overall accuracy (GPT-4o-mini):** MedGraph +10.5 pp above best baseline (Zep).
- **Overall accuracy (Qwen2.5-72B):** MedGraph +10.4 pp above best baseline.
- **Outdated-state errors:** Reduced by 68% relative to Zep.
- **Safety-fact recall:** 100% (by mandatory prepending) vs. 73–82% for retrieval-based baselines.
- **Cross-patient contamination rate:** 0% (by graph isolation) vs. 3–7% for vector-memory baselines.

Ablations and Interpretation

Without mandatory safety block: Safety-fact recall drops from 100% to 79%, confirming retrieval alone is insufficient.

Without typed temporal updates: Using simple entity overwrite increases outdated-state errors 3.2×.

Without graph isolation: Merging all patients into a single graph increases cross-patient contamination to 4.1%.

L1 retrieval necessity: Removing raw transcript retrieval reduces overall accuracy by 4.3 pp, showing verbatim historical passages provide useful evidence beyond the structured graph.

Reference

Donghua Zhang, Zizheng Wang, Ze Dong, Yiding Feng, Siyu Qian. **MedGraph: Maintaining Patient State with Graph Memory for Long-Term Medical Dialogue.** arXiv:2611.08433. arxiv.org/abs/2611.08433